Data Science & AI

DATA STRUCTURES Through Python

Linked Lists

Lecture No.- 02

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Recap of Previous Lecture

- Linked List ?
  - Types of Linked List
  - Create a Node in SLL
  - Construct SLL
  - Search operation
  - Traverse/Access operation

Topics to be Covered

Insertion into SLL

- Insertion at Beginning
- Insertion at End
- Insertion at middle

Topic : SLL - Insertion

```
class Node_Creation :
    def __init__(self, data):
        self.data = data
        self.next = None

class head_Node :
    def __init__(self) :
        self.head = None
```

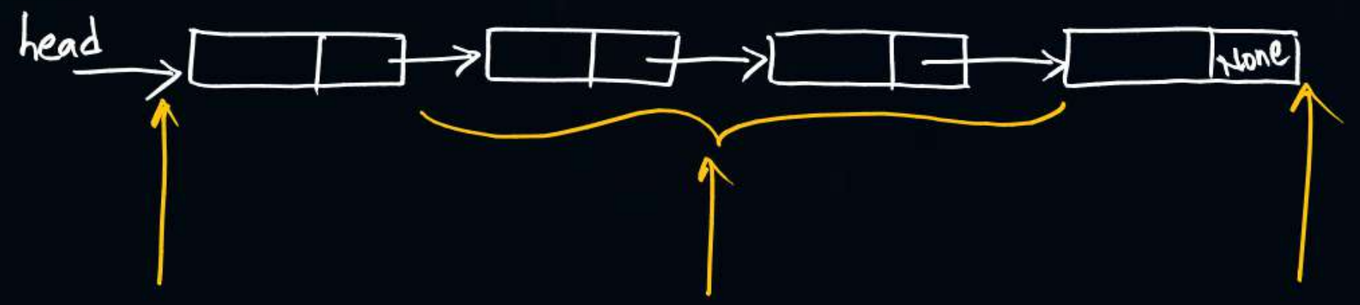
```
class Construction(Node_Creation, head_Node):
    def Construction_SLL:
        New_Node = Node_Creation(10)

        if self.head is None :
            :
            :
            :
            :

Construction_SLL('A')
Construction_SLL('B')
            :
            :
            :
```

Topic : SLL - Insertion

```
def search_SLL(key):

    temp = self.head
    status = 0
    while temp.next is not None:

        if key == temp.data
            Print("key is found")
            status = 1
            break

    if (temp.data == key) and status == 0:
        Print("key is found")
        status = 1
    elif status == 0:
        Print("key is Not found")
```

Topic : SLL - Insertion

Insertion into SLL :



1) Insertion at the front (or) beginning of SLL

2) Insertion at the middle of SLL

3) Insertion at the End of SLL

Topic : SLL - Insertion

Insertion at Beginning

Case-1

only head, No Tail reference

Time Complexity : $O(1)$

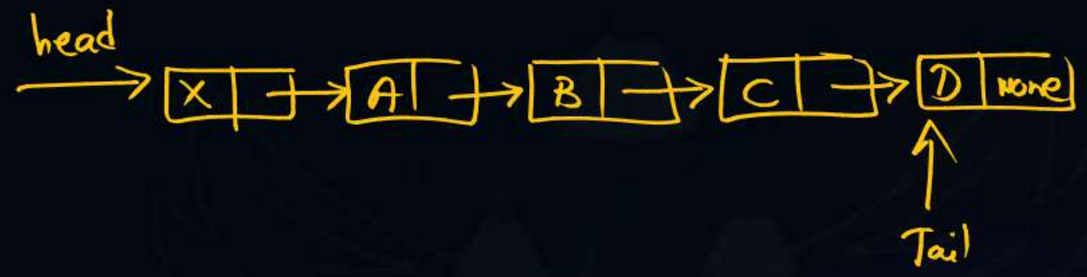
```
def insert_front(Value)   # value = X
    New_Node = Node_Creation(value)
    if self.head is None :   # Empty list
        Self.head = New_Node
        New_Node.next = None
    else:
        New_Node.next = Self.head ✓
        Self.head = New_Node
```

Case-2

With head at first Node, Tail at last Node

- Insertion_front with (or) without Tail has No impact

So, Time Complexity == $O(1)$



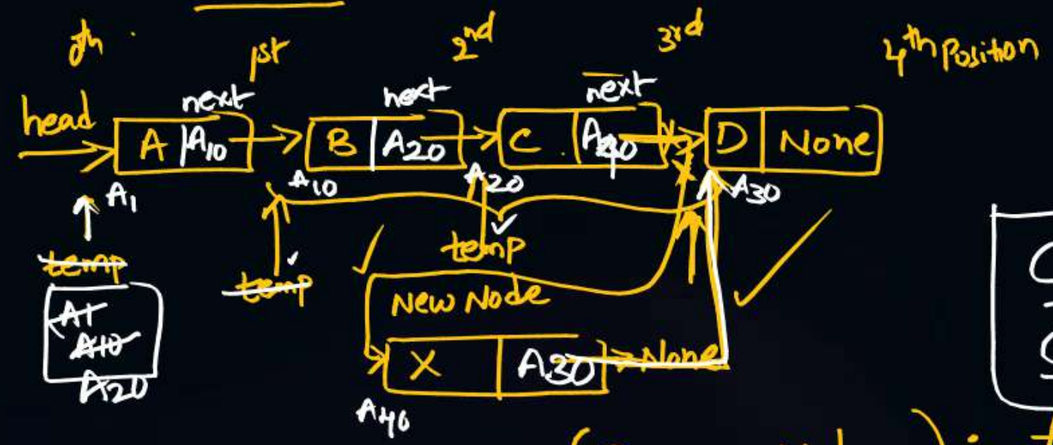
After Insertion of 'X'

Topic : SLL - Insertion

Insertion_At_Middle [Neither at front, Nor at last]

Worst Case : Inserting a Node at Second Last Position

Case_1 (without Tail)



After Insertion ⇒

Case_2 with Tail at last Node has No impact : $O(n)$

Case_3 : Object at required Node ['ob' at 'C' Node] $= O(1)$

Time Complexity

```
def Insert_middle(Position, Value):   # Position=3, value='X'
    count = 1
    New_Node = Node_creation(Value)
    if Self.head is None:
        self.head = New_Node
        New_Node.next = None
    else:
        temp = self.head
        while count < Position:   1<3 True, 2<3 True, 3<3 False
            temp = temp.next
            count = count+1
```

Inside else only, after while

```
NewNode.next = temp.next
temp.next = New_Node
```

Best Case : $O(1) = \Omega(1)$

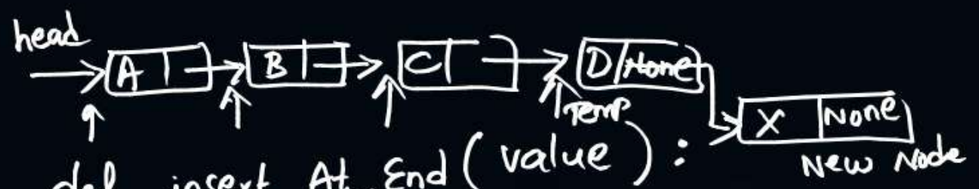
Avg Case : $O(\frac{n}{2}) \Rightarrow O(n)$
$\Rightarrow \Theta(n)$

Worst Case : $O(n-1)$
$\cong O(n)$

Topic : SLL - Insertion

Insertion_At_End

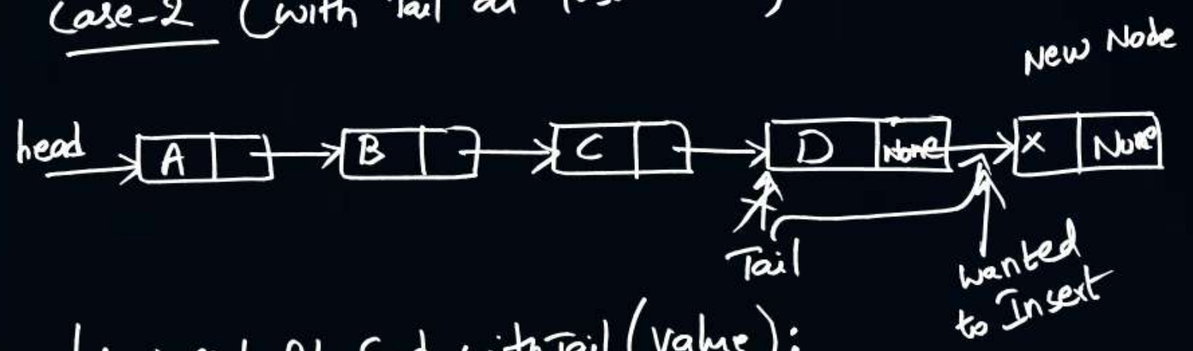
Case-1 (without Tail)




```
def insert_At_End(value):
    New_Node = Node_Creation(value)
    if self.head is None:
        self.head = New_Node
        New_Node.next = None
    else: temp = self.head
        while temp.next is not None:
            temp = temp.next
        temp.next = New_Node
        New_Node.next = None
```

Time Complexity
$O(n)$

Case-2 (with Tail at last Node)




```
def insert_At_End_withTail(value):
    New_Node = Node_Creation(value)
    if self.head is None:
        self.head = New_Node
        self.Tail = New_Node
        New_Node.Next = None
    else:
        Tail.next = New_Node
        New Node_Next = None
        Tail = New_Node
```

Time Complexity
$O(1)$

2 mins Summary

- SLL_insertion operation
  - At Beginning
  - At End
  - At given Position/middle

  } with Time Complexities and Pseudo Code.

To be Contd...

THANK - YOU